



Service Procedure No.: 05-03-05
Service Procedure Title: MCP Removal and Replacement: 5 nm Column
Applicable Systems: 8000, Micrion 9800 FlipChip, MicroMill HT
Frequency of Service: As needed.

Introduction

This procedure describes the removal of the microchannel plate (MCP) from the column of the FIB system, and the installation of a replacement MCP.

Materials Required

This procedure requires the use of the following materials:

- Clean, standard field service kit
- Lint-free wipes
- Clean-room gloves
- Laboratory-grade isopropyl alcohol

NOTICE

Per paragraph 11.1 of Safety Guidelines for Semiconductor Manufacturing Equipment (S2-93), this procedure is a Type 5 task. Specifically:
“Equipment is energized and measurements and adjustments require physical entry into the equipment, or equipment configuration will not allow the use of clamp-on probes.”

Part 1. System Preparation

Configure the Chemical Sources in “Safe” Mode

Before performing the tasks described in this procedure, you must put the chemical sources in a “safe” mode; see Service Procedure 02-01.

WARNING

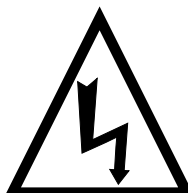


Do not proceed until the chemical sources are secure. The chemical reservoirs in FIB systems contain hazardous materials. Labels affixed to the exterior of a chemical cabinet identify the chemical(s) contained inside. If a chemical source is not in its “safest” state, as described in **Service Procedure 02-01**, it is possible for a combination of component failures to release the chemical(s) through a cabinet’s delivery line, a roughing line, a purging line, or from a chemical reservoir into the workchamber. Do not open the workchamber, or disassemble any part of the vacuum pumping system, or detach any gas line from any chemical cabinet until you are sure that every chemical source attached to the FIB system is in its “safest” state.

Power Down and Vent the System

Before performing the tasks described in this procedure, you must power down and vent the system; see Service Procedure 01-02.

WARNING



Dangerously high voltages are present in the process module while the FIB system is powered up. Do not rely *only* on the software Power Down function. Disable the voltages using both manual and automatic procedures, as described in **Service Procedure 01-02**. Do not proceed until you have switched off the applicable power supplies.

Clean off the exterior of the system as well as the floor; see Service Procedure 06-05.

Open the Workchamber

Remove the front panels to gain access to the workchamber door.

Once the system is completely vented, open the workchamber door by removing the 1/2” socket head cap screws.

WEAR GLOVES, DO NOT HANDLE SOLVENTS

When working inside the chamber or when handling any components that will be exposed to vacuum, wear gloves rated for Class 100 environments or better. Due to disintegration of such gloves by solvents, do not handle solvents or parts that are wet with solvent.

Part 2. Removing the MCP

- 2-1. After the system has vented, open the workchamber. Two types of flood gun/MCP configurations are installed in FIB systems with 5 nm columns: a flood gun without a snout combined with a small MCP^[1] (as shown in Figures 1 and 2), and a flood gun with a snout paired with a large MCP^[2] (as shown in Figures 3, 4, and 5).
 - 2-2. Wearing clean room gloves, loosen the flood gun height positioning screw (see Figures 6 and 7) and slide the flood gun assembly down to provide access to the MCP.
 - 2-3. Remove the MCP electrical connectors (see Figures 6 and 7).
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NOTICE: DO NOT DROP MOUNTING SCREWS OR SPACERS

Before removing and replacing the MCP, place a clean lint-free wipe below the MCP to prevent the mounting screws (and spacers, if applicable) from falling into the turbo pump or the stage hardware. **If a screw or spacer is dropped into the workchamber, be sure to retrieve it before proceeding.**

- 2-4. Remove the MCP from the bottom of the column by removing the mounting screws (and spacers, if applicable). See Figures 6 and 7. Be careful not to damage the MCP screen.

Note: to eliminate the risk of dropped hardware, remove the column (as described in Service Procedure 05-09-05), replace the MCP, and then reinstall the column.

[1] There are two styles of small MCP: flat and angled (see Figure 2). Both styles have the same fasteners and electrical connectors.

[2] There are two configurations of the flood gun with a snout and large MCP combination (see Figures 3 and 4). In a 9800 series system (as shown in Figure 4), the flood gun is rotated 90° to accommodate the installation of an optical microscope.

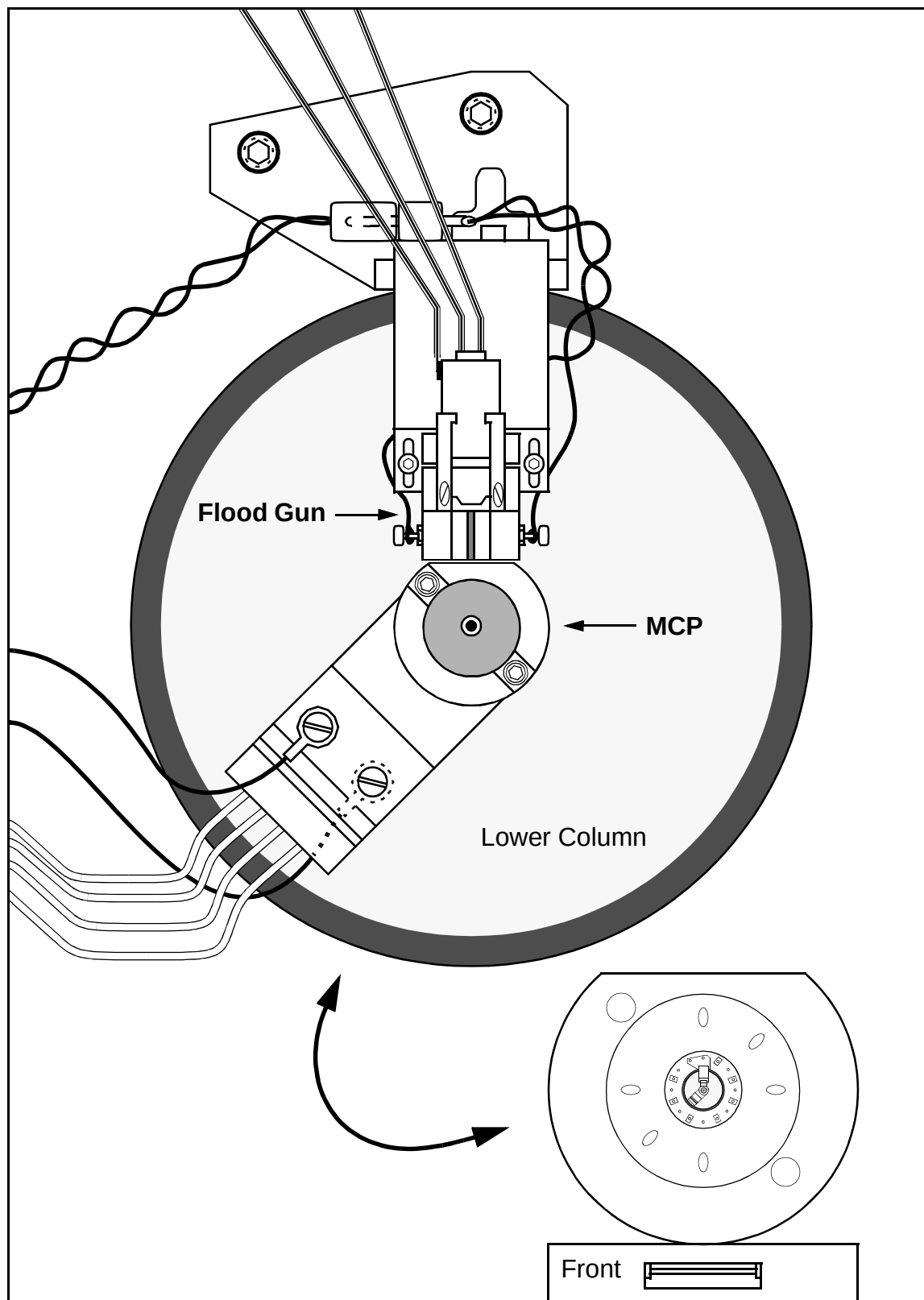


Figure 1. Flood Gun Without Snout and Small MCP: Interior View

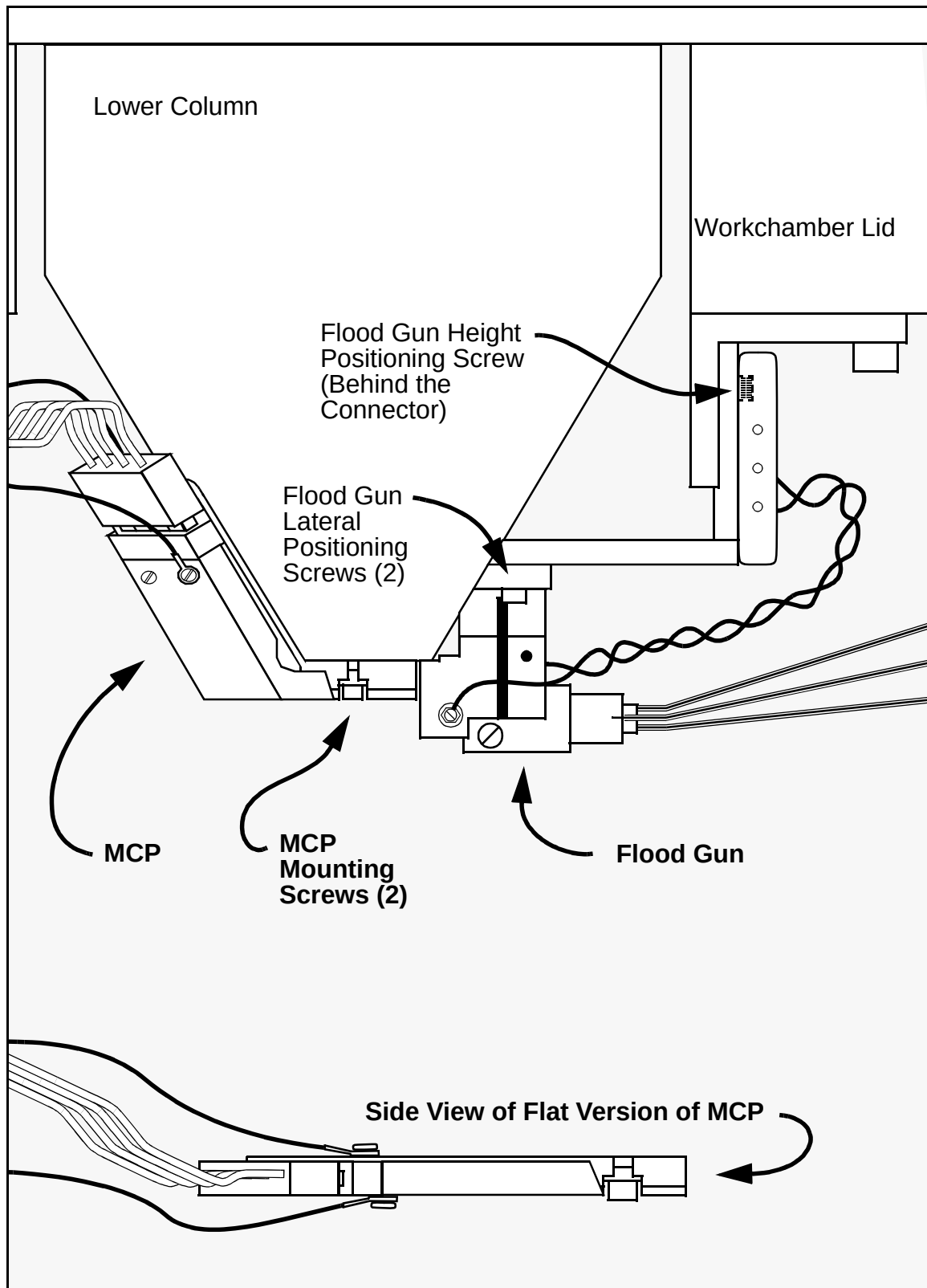


Figure 2. Side View of Flood Gun Without Snout and Small MCP(s)

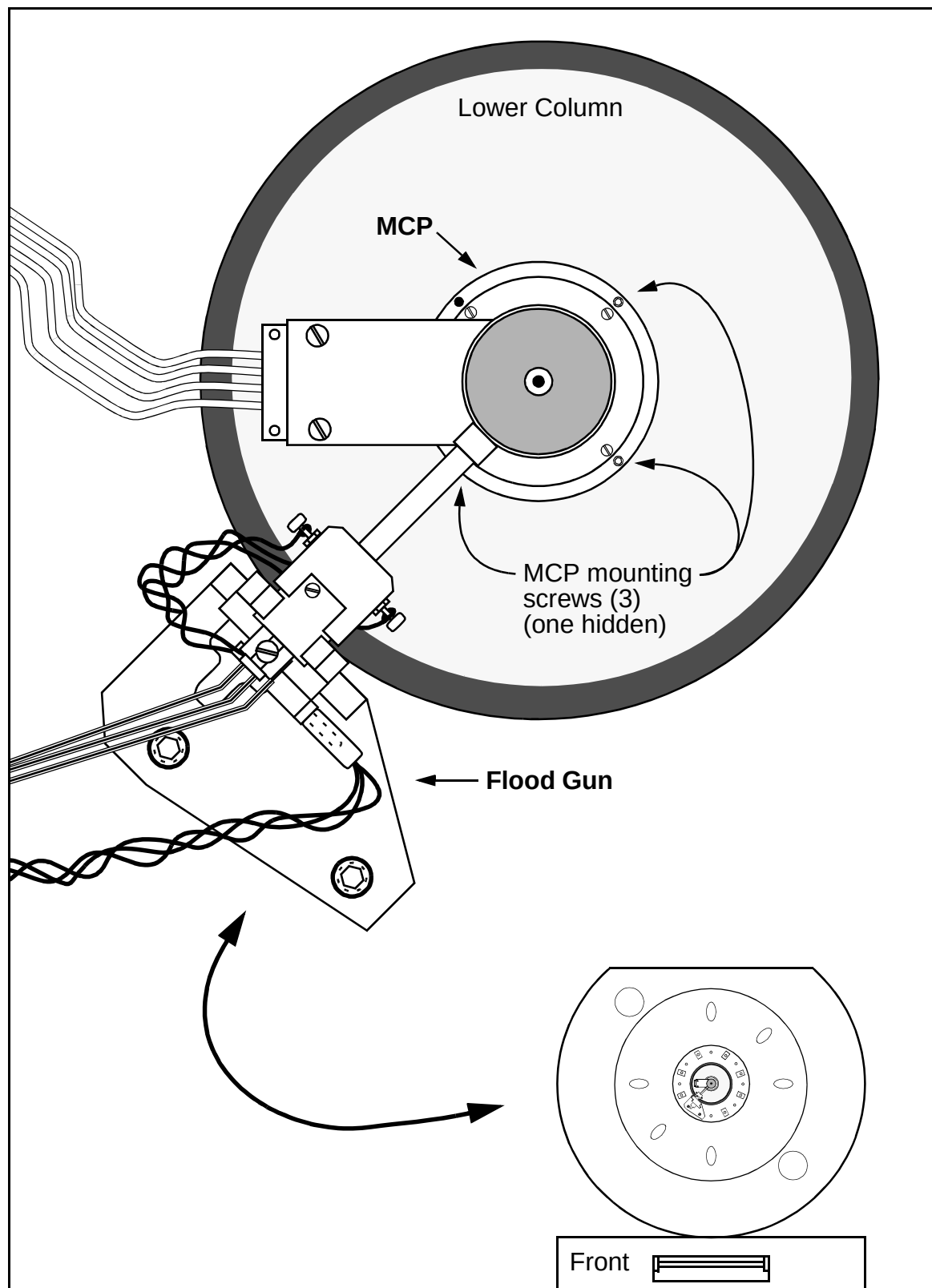


Figure 3. Flood Gun With Snout and Large MCP: Interior View

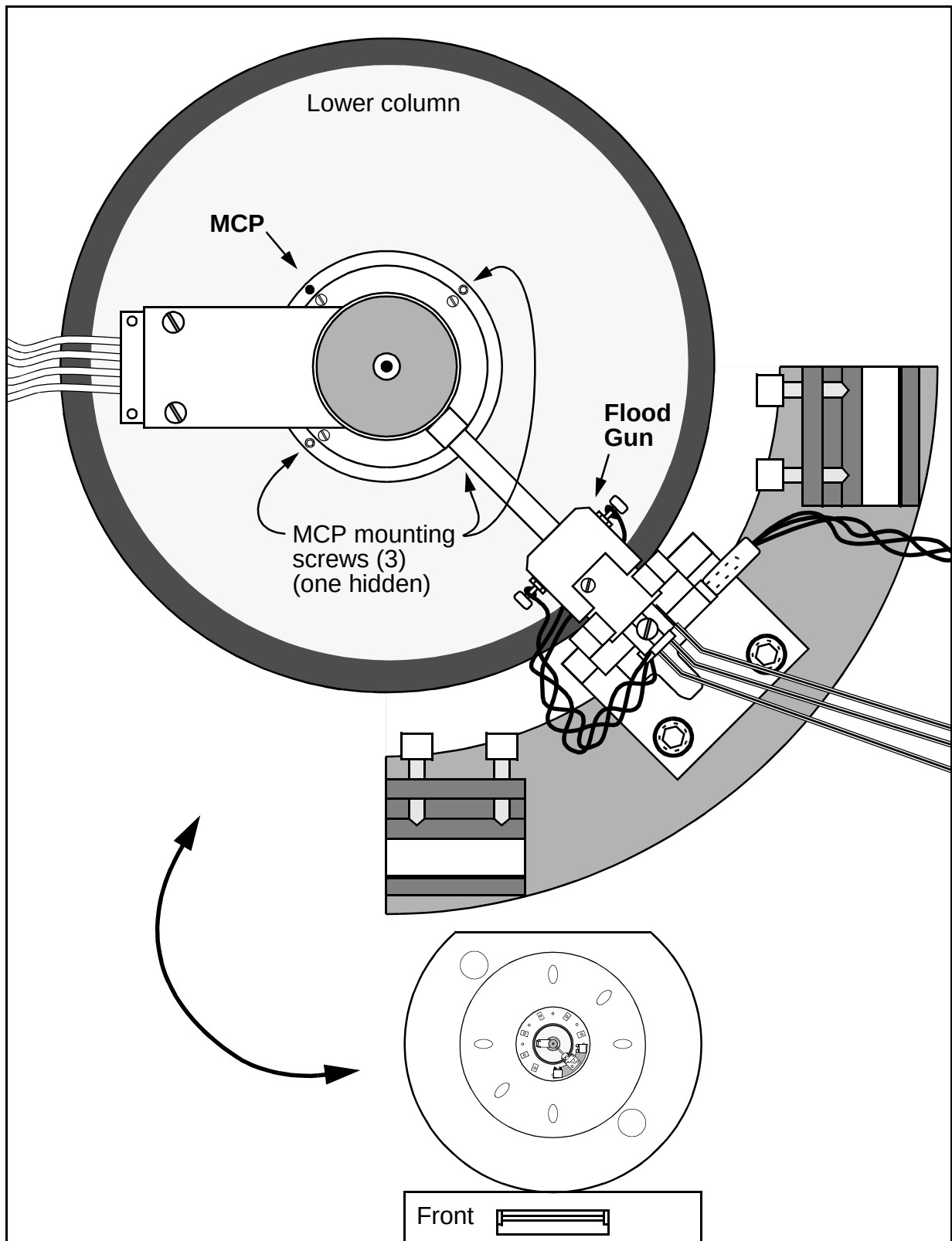


Figure 4. Flood Gun With Snout and Large MCP in 9800 System: Interior View

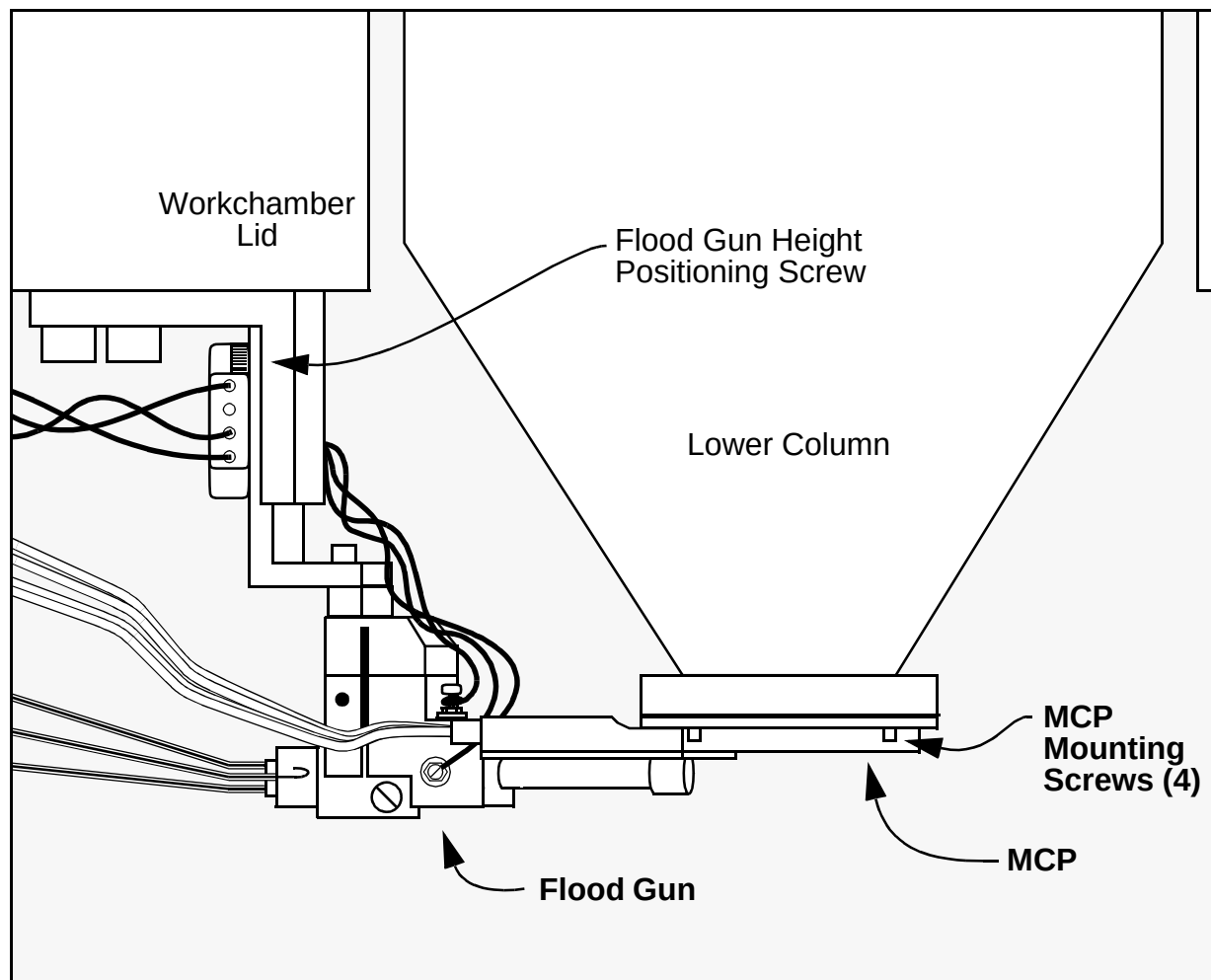


Figure 5. Side View of Flood Gun With Snout and Large MCP

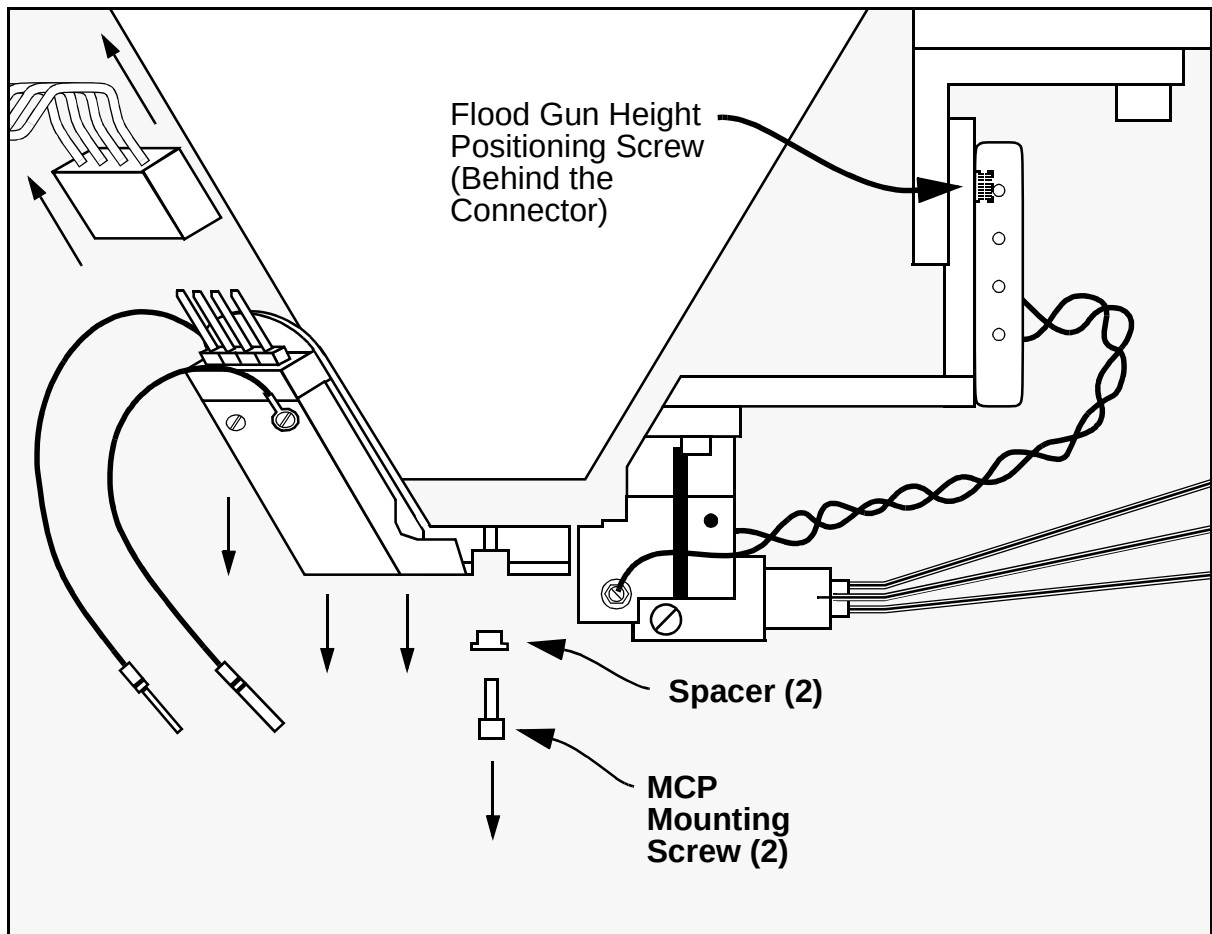


Figure 6. Side View of Flood Gun Without Snout and Small MCP: Flood Gun Lowered and MCP Disconnected.

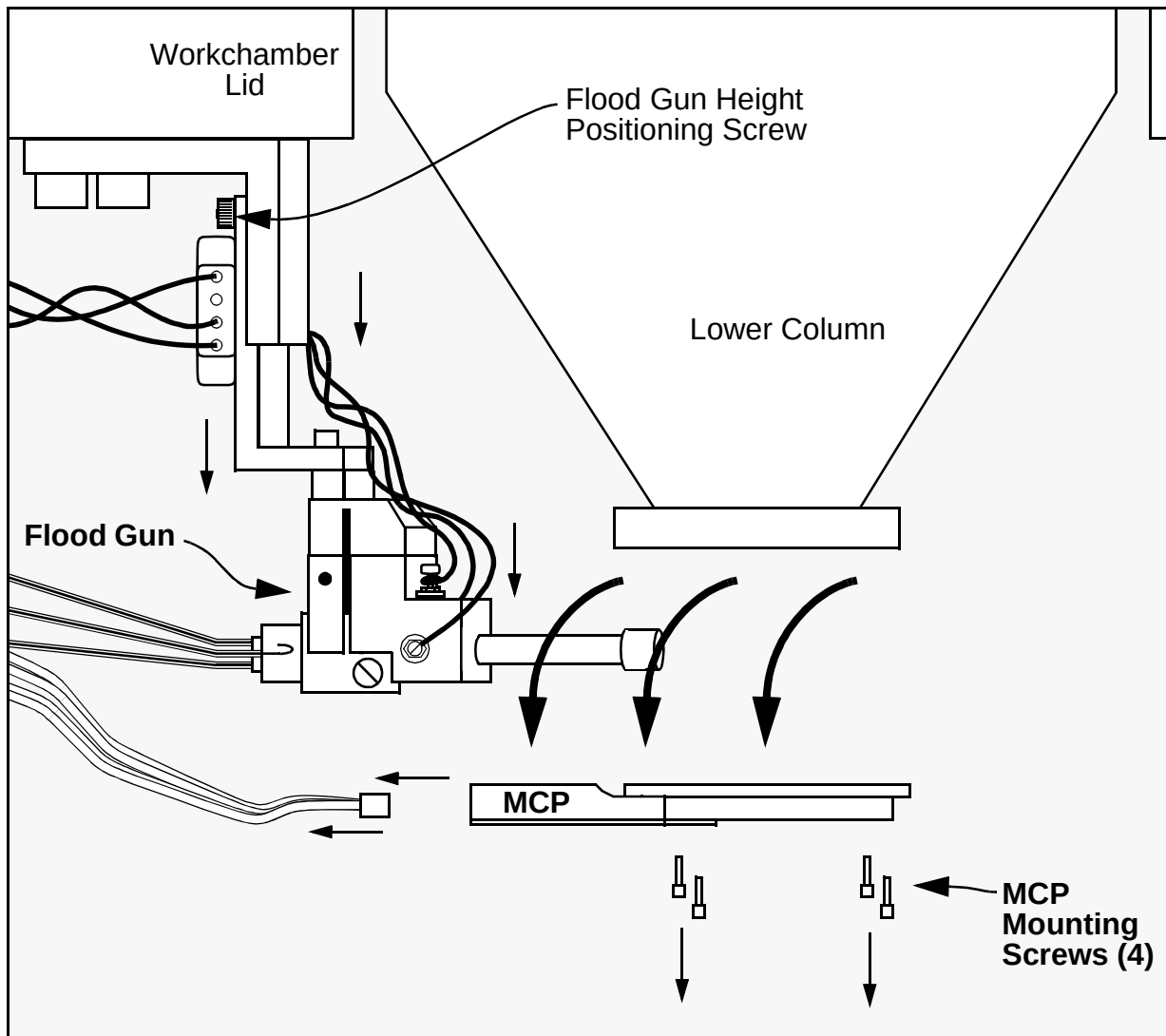


Figure 7. Side View of Flood Gun With Snout and Large MCP: Flood Gun Lowered and MCP Disconnected.

Part 3. Replacing the MCP

- 3-1.** Fasten the replacement MCP to the bottom of the column with the MCP mounting screws (see Figures 6 and 7).
- 3-2.** Reattach the MCP electrical connectors.
- 3-3.** Slide the flood gun subassembly up until it contacts the lower column. Make sure that the flood gun is aligned with (points directly at) the center bore of the MCP (see Figures 1, 2, 3, 4, and 5). Tighten the flood gun height positioning screw (see Figures 2 and 5).
- 3-4.** Clean the workchamber (see Service Procedure 06-03), then close the workchamber door.
- 3-5.** Power up the system as described in Service Procedure 01-01, and pump down the system as described in Service Procedure 04-03.